

TCS-504: System Design — Assignment 1

3. Class Responsibility Collaboration (CRC)

Class	Knows (State)	Does (Behavior)	Must NOT Do
Movie	movieId, title, language, durationMinutes	Exposes read-only getters	Know about shows, screens, or pricing
Seat	seatNumber, type	Reports base seat price	Track availability or booking status
Screen	screenNumber, seats	Owns physical seats; exposes seat list	Know about movie schedules
Cinema	name, movies, screens	Adds/lists movies and screens	Handle bookings or payments
Show	showId, movie, screen, startTime, showSeats	Prints seat matrix; looks up ShowSeat	Process payments or create bookings
ShowSeat	seat (Seat*), status (AVAILABLE/BOOKED)	Flips status (book/release)	Know about customer or pricing
Customer	name, phone	Provides user identification	Manage bookings directly
Booking	bookingId, show, customer, seats, total, status	Transitions status (confirm/fail/cancel)	Print ticket or handle payment gateways
Payment	amount	Abstract pay(amount) interface	Directly alter booking records
BookingService	bookings, nextBookingId	Orchestrates booking, payment, cancellation	Run CLI UI prompts
PriceCalculator	None (Stateless)	Calculates total price from ShowSeats	Mutate seat availability
TicketPrinter	None (Stateless)	Formats and outputs ticket to console	Mutate booking status